

University of Waterloo  
ECE 457B: Fundamentals of Computational Intelligence  
Winter 2025  
Assignment 2 - Representation Learning, Parametric Regression  
Algorithms  
Submission Due: March 9, 2025 by 11:59pm

## Overview

**Collaboration/Groups:** You may do your work individually or with a partner. If you are working with a partner, you must sign up for an **Assignment Group** in LEARN and Crowdmark. If you are working alone, you should still join a group in LEARN with one member. You can also collaborate with other classmates on the right tools to use and setting up your programming environment, but your submitted work must be only from members of your group.

**Submission:** Hand in one report per person, or group, to Crowdmark and LEARN. Your report should be submitted as two files:

- *To your group dropbox on LEARN:* Your code in the form of a jupyter notebook that has the code, and results already generated on the provided data in a readable way.
- *To Crowdmark:* Your report in the form of a pdf that is *no more than about 10 pages*. You can use the jupyter notebook to generate this report document, but you need to **remove all the code and preliminary processing** from it. Only keep what is actually essential to explain what you did, why you did it and what your results show. It might be easier to copy the text, headings, plots and tables out into a separate document and save that.
- For the theory questions, you can write the answer out by hand on paper or a tablet and take a picture or save to pdf/image. You could also choose to type it out, Microsoft Word and Google Docs now have quite good math notation entry, or you could use the best (but not the easiest) option LaTeX using the Overleaf online editor (<https://www.overleaf.com/edu/uwaterloo>). UWaterloo students get free access.

Your group on LEARN has an associated **dropbox**.

## Presentation of Results (Descriptive, Concise, Clear)

- **Report document:** All tasks and questions below should be carried out or answered for both datasets. The report is a pdf document with your answers for the questions, description of your process, plots and tables for results. *It does not contain code!*

- **Essential Plots and Tables:** Along the way, minimize the number of different plots shown to the *essentials* for the reader to understand what you did.
- **Summary Table:** At the very end you will produce a summary table of all the accuracy results for the different experiments.
- **Code:** You must submit your code, in a self-contained python jupyter notebook to LEARN. Your, you should collapse most of your code before printing to improve readability. You only need to show critical code which relates to the central task of the question or a point you are highlighting in your text.

**Tools:** You can use libraries available for python which we discuss in class or provide as links. You need to mention explicitly which libraries you are using, any blogs or papers you used to figure out how to carry out your calculations.

#### Specific objectives:

- Load datasets, run simple classification/regression algorithms and generate some exploratory plots and tables. Ensure all plots and tables have labels for axis, titles and short captions explaining them.
- Practice how to apply the methods discussed in class.
- Demonstrate understanding by making well reasoned design choices, and *explaining* any result you obtain in straightforward, short, text. **Remember**, there isn't always one particular right answer to how to do something, but you must justify what you did and demonstrate how it worked.

## Dataset

To keep things simple, for this assignment we will re-use datasets from asg1, since you've already preprocessed those and have classification results for some algorithms, you can reuse some of that code and setup.

In addition, we'll add one more dataset **forest fire** from the same UCI repository. This is an old dataset, but later in the course we'll try to look at a more recent dataset used our research with the same problem.

The first data set is the Wine Quality Data Set:

- Wine Quality Dataset:  
<https://archive.ics.uci.edu/ml/datasets/wine+quality>
- The dataset original consists of one datafile for red wine and one datafile for white wine. The provided jupyter notebook loads the data and adds an additional column for wine `color` where `color = 0` indicates white wine and `color = 1` indicates red wine.

- **Regression task:** predict ‘quality’ from the other features, including ‘color’.

The second dataset is the Abalone dataset about the abalone fish and its physical characteristics.

- An `abalone.csv` file will be available in the `asg1` directory on learn.
- Original Dataset:  
<https://archive.ics.uci.edu/dataset/1/abalone>
- **Regression task:** Now we’ll predict the ‘Rings’ value from the other features as continuous number and measure our error from the true label.

(NEW!) An additional dataset to look at is about forest fires. This dataset has some challenging scale issues, there are a large number of very small fires and small number of large ones. The dataset description page has some tips of how it was transformed and analysed by the original researchers, you can try that or something else.

- The `forest fires` dataset can be obtained online at the Dataset url.
- Dataset:  
<https://archive.ics.uci.edu/dataset/162/forest+fires>
- **Regression Task:** predict the ‘Area’ target from the other features.

## 1 MLE and MAP Derivation

In this problem we will find the *maximum likelihood estimator (MLE)* and *maximum a-posteriori (MAP)* estimator for given posterior and prior distributions. We assume we have  $N$  samples,  $x_1, \dots, x_i, \dots, x_N$ , each *independently* drawn from a *Kumaraswamy distribution*, defined as :

$$x_i \sim \text{Kumaraswamy}(a, b) : f(x|a, b) = abx^{a-1}(1 - x^a)^{(b-1)} \quad (1)$$

where  $x_i \in (0, 1)$ . The parameters  $a, b$  are positive real numbers and act as *shape parameters* in a similar way to the *Beta distribution*. For the purposes of this question we assume that  $a$  is known, and we need to derive a formula for estimating  $b$  from data.

**Submission:** For this question you can submit your answer in any way which is clear to read and understand. You can type it up in  $\text{\LaTeX}$  or some other tools that lets you type math and submit as a pdf. Alternatively, you can also submit a hand written solution from a your tablet or paper. In this case, be sure to write *very neatly*, then export the image or take a picture, of each page, and submit these images as your answer for the question.

1. (15 points) Derive the **MLE estimator** for  $b$ . Make sure to show all of your work.  
Steps for MLE:

- (a) Define the form of the likelihood term as the posterior formed by the product of distribution from (1). For defining the log-likelihood, we can assume that  $\log(x) = \ln(x)$ .
  - (b) Take the log of the likelihood.
  - (c) Take the derivative of the log-likelihood and set this equal to zero
  - (d) Solve for the target parameter.
2. **(15 points)** Derive the **MAP Estimator** estimator for the same distribution. Derive the MAP estimator for  $b$ . We assume that as a prior distribution that  $b$  is itself sampled from a *Gaussian distribution* with known mean  $\mu$  and known variance  $\sigma^2$ .

$$b \sim \mathcal{N}(\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(b-\mu)^2}{2\sigma^2}} \quad (2)$$

Steps for MAP:

- (a) As before, define the form of the log-posterior term from the given posterior distribution now including the given prior distribution.
- (b) Take the derivative of the log-likelihood and set to equal to zero.

**Note:** For this part, a clean analytical form for  $b$  will not be possible. Derive the formula for  $\text{MAP}_a$  in general terms using the known distribution's and reduce for  $b$  as much as you can.

## 2 Regression

**(35 points)** Using a similar approach to assignment 1, train a number of models on the three datasets for the regression task of **predicting the numeric target variable** and measure your performance using, at least, **root mean squared error (RMSE)**. You should still use 5-fold cross validation as before.

Use your judgment and experience to decide how to train these algorithms and choose their settings. At a minimum, pick some good hyper-parameter settings, train the models, and show some analysis of their performance measured by **RMSE** compared to each other. You can attempt to tune the hyper-parameters once again for these tasks or you can stick to the best values you found in the assignment 1.

As always, justify whichever approach you use for processing or data, design of your training, and choices of algorithms/hyper-parameters. Highlight best/worst performance. Be sure to consider the *mean and variance* results given the k-fold cross validation.

Algorithms to use:

- $k$ -Nearest Neighbours (`KNeighborsRegressor`)
- (optional) Decision Trees (`DecisionTreeRegressor`)

- RandomForests (`RandomForestRegressor`)
- Gradient Tree Boosting (`GradientBoostingRegressor`, or some other implementations if you have a preference).
- (optional) feel free to add some other variants of these or algorithms related to the ones we've talked about so far.

### 3 Representation Learning

**(35 points)** Apply PCA, t-SNE, and one other feature extraction method on the datasets, analyse the resulting new representations in terms of interpretability (using plots) and prediction impact (by training models for regression).

1. Use the t-SNE method to visualize the datasets by producing a 2D plot, comment on any useful patterns that this shows.
2. Run PCA on each dataset, look at the total variance explained by the principal components. Plot the first two principle components (you could even compare multiple pairs of components), using easily distinguishable colours and markers to indicate the labels of each datapoint. Comment on any interesting patterns you find.
3. Repeat the previous step with some other feature extraction method (I'll refer to it as TBD method of your choice. It could be anything from `sklearn.decomposition`, or LDA a method from `sklearnr.discriminant.analysis`, or a nonlinear method from `sklearn.manifold`, or *something else I haven't seen?*)
4. Create a **scree-plot** to look at the cumulative variance represented by the PCA eigenvectors. Use this and any other analysis you choose to determine a reduced dimensionality size. Create two new versions of your datasets (eg. `wine-pca`, `wine-tbd`) using the best reduced dimensionality representations from PCA and TBD.
5. Rerun your regression experiments on the new reduced datasets and compare the performance.

Include summary performance measures (at least RMSE) for all nine dataset combinations in a table and comment on any overall patterns.

| Dataset (orig, -pca, -tbd) | k-NN | RF | GB | ??? |
|----------------------------|------|----|----|-----|
| wine                       |      |    |    |     |
| wine-pca                   |      |    |    |     |
| wine-tbd                   |      |    |    |     |
| ...                        |      |    |    |     |